

Article

Eroding the First Rung: Generative Artificial Intelligence, Organizational Learning, and the Self-Reinforcing Decline of Youth Employment in Chinese Listed Firms

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Abstract

Entry-level work is the first rung of the workforce development ladder, and generative artificial intelligence is now most capable at precisely the tasks that rung is made of. This study examines how the adoption of generative artificial intelligence by firms reshapes the youth employment share, and asks whether the effect is a one-way shock or a self-reinforcing loop within the firm's socio-technical system of skill formation. The analysis uses 25,701 firm-year observations for 3,000 Chinese A-share listed firms over 2016–2024, with generative-AI application intensity measured from annual-report text. A two-way fixed-effects panel model is estimated as the benchmark, three-step mediation with firm-block bootstrapping identifies the transmission channels, instrumental variables, propensity score matching and entropy balancing address endogeneity, and a bias-corrected panel vector autoregression with orthogonalized impulse responses tests the feedback structure. A one-standard-deviation increase in adoption lowers the youth employment share by 1.74 percentage points, or 6.0 percent of its mean, while employment of workers aged 31 to 50 is barely affected. Falling training investment and a contracting routine task base jointly account for 34.9 percent of the total effect. The effect is weaker where organizational learning capability is stronger and sharper where relative labor costs are higher. Critically, the erosion feeds back: weaker training and a smaller youth share raise subsequent adoption, and this reinforcing loop amplifies the eight-year cumulative displacement by 19 percent. Policy should therefore target the loop rather than the shock.

Keywords: generative artificial intelligence; youth employment; organizational learning; socio-technical systems; feedback loop; capability trap; workforce development; panel vector autoregression

1. Introduction

Youth employment has become one of the most closely watched indicators of economic health in the world's large economies, and it is deteriorating faster than any other segment of the labor market. The International Labour Organization reports that young people constitute roughly one-sixth of the global working-age population but close to two-fifths of the unemployed, and that their labor force participation fell more than three times as fast as that of the working-age population as a whole over the past decade [1]. In China the pressure is acute. The number of new college graduates reached 12.22 million in 2025, the highest on record [2], while the surveyed urban unemployment rate of the population aged 16 to 24 excluding students remained above 17 percent through much of the year, more than three times the national rate [3]. The scale of the cohort and the persistence of the gap have made the absorption of young workers a central objective of industrial and employment policy [4,5]. What is at stake is not only the current year of joblessness. Cohorts that enter the labor market in slack conditions carry measurable earnings and employment penalties for a decade or more, and the penalty is largest for those who never secure a first foothold [6–9].

Over the same period the technological environment facing firms changed abruptly. Generative artificial intelligence, which produces language, code and images rather than merely classifying them, moved from laboratory demonstration to routine workplace tool within two years of the release of large conversational models [10]. Its capability profile is unusual. Field and experimental evidence shows that generative models raise the productivity of writing, customer support, consulting and software work substantially, and that the gains are largest for the least experienced workers, whose output moves closest to that of their more senior colleagues [11–14]. Task-level exposure measures point in the same direction: the occupations most exposed are cognitive and white-collar rather than manual, and within them the most exposed tasks are the codifiable, supervised ones that junior staff are normally hired to perform [15,16]. Early aggregate evidence is consistent with a distributional consequence. Employment of workers aged 22 to 25 in the most exposed occupations has declined relative to that of experienced workers in the same occupations, while senior employment has not [17,18].

Existing scholarship reads this as a labor demand problem. In the task-based framework, a technology that automates a set of tasks displaces the labor previously allocated to them and, where it also creates new tasks, reinstates demand elsewhere [19–21]. Because generative models are a general-purpose technology whose complementary investments arrive with a lag, the composition of the adjustment may differ sharply from its eventual level [22–24]. Applied to generative artificial intelligence, this yields a clear and testable prediction about the level of entry-level hiring, and firm-level studies in several countries have begun to confirm it [25–27]. Yet the framework treats the firm's technology choice as exogenous to the composition of its workforce, and treats each year's adjustment as

independent of the last. Neither assumption is innocuous when the resource being displaced is the input to the firm's own future capability.

Entry-level positions are not only jobs; they are the mechanism by which firms manufacture the experienced workers they will need later. Human capital theory and the literature on internal labor markets describe a ladder along which workers acquire task-specific and firm-specific skill through supervised practice, and along which the firm recovers its training outlay in later periods [28–31]. Organization theory adds that this ladder is also the firm's absorptive capacity: the stock of people able to recognize, assimilate and apply new external knowledge [32,33]. When a technology substitutes for the bottom rung, the firm does not simply buy the same output more cheaply. It stops producing an input it cannot buy on the market, and it does so in a period in which the consequence is invisible because the existing stock of experienced staff still suffices.

This is the structure that system dynamics calls a capability trap. Repenning and Sterman [34] showed that when a short-run fix and a long-run capability investment compete for the same resources, and when the returns to the investment are delayed, organizations reliably choose the fix, thereby degrading the capability that would have reduced the need for the fix, and so raising the pull toward the fix again in the next period. Rahmandad and Repenning [35] formalized the resulting erosion dynamics; Levinthal and March [36] described the same pathology as the myopia of learning. The behavior is not irrational at any single decision point. It is the emergent property of a reinforcing loop with a delay, and it is invisible to any analysis that estimates one arrow of the loop at a time. This is the sense in which the firm is a complex adaptive system rather than a production function: its behavior over time is generated by the interaction of its parts, not by any of them separately [37,38]. Socio-technical systems theory makes the same structural claim in different language: a technical subsystem and a social subsystem are jointly optimized or jointly degraded, and an intervention in one that ignores the other produces outcomes that neither designer intended [39,40].

Although existing studies have thoroughly documented the exposure of young workers to generative artificial intelligence and the productivity gains the technology delivers, they have predominantly treated the relationship as a unidirectional shock running from technology to employment, neglecting the possibility that the employment response itself changes the firm's subsequent technology choice. Three gaps follow. First, the mechanisms are asserted rather than measured: displacement is attributed to task substitution without a test of whether the firm's investment in skill formation also falls. Second, the moderating role of the firm's own learning capability is rarely examined, even though it is the most obvious leverage point for management. Third, and most importantly, no firm-level study has estimated the return arc of the loop, so the literature cannot say whether the observed displacement is a level shift that will exhaust itself or a self-reinforcing process that accumulates.

Based on this, this study uses a sample of Chinese A-share listed firms over 2016–2024 to examine how generative artificial intelligence adoption reshapes the youth employment share, why it does so, what conditions the effect, and whether it feeds back on itself. Adoption intensity is measured from annual-report text; the youth employment share is taken from the disclosed age structure of the workforce; training expenditure per employee and the

routine task base serve as the two mediators; research and development intensity and relative labor cost serve as the two moderators. The benchmark is a two-way fixed-effects panel model. Mechanisms are identified by three-step mediation with firm-block bootstrapped indirect effects. Endogeneity is addressed by an overidentified instrumental-variable design, propensity score matching and entropy balancing. The feedback structure is estimated by a bias-corrected panel vector autoregression with Granger causality tests, a stability check and orthogonalized impulse responses, from which the amplification attributable to the loop is computed by comparing the closed system with a counterfactual system in which the return arcs are switched off.

Accordingly, this study makes marginal contributions in the following aspects. First, it moves the analysis of generative artificial intelligence and youth employment from a unidirectional treatment effect to a feedback structure, and provides what appears to be the first firm-level estimate of the return arc that closes the loop, together with a quantification of how much the loop adds to the cumulative displacement. Second, it opens the mechanism, showing that the erosion of the firm's skill-formation investment is a distinct and measurable channel alongside the task-substitution channel that the task-based framework emphasizes, and thereby connects the labor economics of automation to the organizational learning literature. Third, it identifies organizational learning capability as a leverage point in the systems sense: the same technology, adopted at the same intensity, displaces markedly less youth employment in firms that have built the capacity to absorb it, which turns an apparently technological outcome into a manageable one. Fourth, it supplies evidence from the largest developing labor market at a moment when the policy question is live, and translates the estimates into leverage points that firms and regulators can act on.

The remainder of this paper proceeds as follows. Section 2 reviews the related literature and develops the hypotheses. Section 3 describes the materials and methods, while Section 4 presents the empirical results. Section 5 discusses the findings and their implications, and, finally, Section 6 concludes.

2. Literature Review and Hypothesis Development

2.1. Generative Artificial Intelligence and the Demand for Entry-Level Labor

The task-based framework treats production as a continuum of tasks allocated between capital and labor, so that a technology which automates a range of tasks reduces the labor demanded at given wages unless it simultaneously creates new tasks in which labor holds the comparative advantage [19,20,41]. The framework has organized a large body of evidence on earlier waves of automation. Ref. [42] found that each additional industrial robot per thousand workers reduced the employment-to-population ratio in exposed United States commuting zones, and Ref. [43] documented similar firm-level adjustments following robot imports. Ref. [44] established the prior result on which this literature rests, namely that computerization substitutes for routine tasks and complements non-routine analytic ones.

Generative artificial intelligence inverts part of that pattern. Ref. [15] estimated that roughly four-fifths of the United States workforce could have at least a tenth of their work tasks affected by large language models, and that exposure rises rather than falls with education and wages. Ref. [16] reached a

comparable ranking of occupational exposure using a different mapping of model capabilities to work activities. What the exposed tasks share is not manual repetition but codifiability: drafting standard correspondence, summarizing documents, writing boilerplate code, conducting preliminary research and preparing first-pass analysis. These are the tasks that firms customarily assign to new entrants, partly because they are valuable and partly because performing them under supervision is how entrants learn [45,46].

The productivity evidence sharpens the implication. Ref. [11] found that a generative assistant deployed to more than five thousand customer support agents raised resolutions per hour by about fourteen percent, with gains concentrated among novice and low-skilled agents and no measurable gain for the most experienced. Ref. [12] and Ref. [13] reported the same compression of the performance distribution in writing and consulting tasks, and Ref. [14] in software development; Refs. [47] and [48] report the same compression in three field experiments with software developers and among Kenyan entrepreneurs respectively. A technology that raises the output of a novice toward that of an expert lowers the price the firm is willing to pay for novices, because what the firm was purchasing was partly the option on the expert the novice would become [49,50]. Consistent with this reading, Ref. [17] documented a relative decline in employment among workers aged 22 to 25 in the most exposed occupations after generative models diffused, with no corresponding decline for experienced workers, and Ref. [18] found weakening entry-level vacancy postings in exposed occupations across several developing economies. Firm-level studies of adoption report reduced junior hiring without measurable effects on senior headcount [25–27]. The asymmetry is not new to generative models: Ref. [51] showed that firms adjust to adverse conditions primarily by suspending entry-level recruitment rather than by dismissing incumbents, so the entry port is the margin that moves first. Work in this journal has begun to trace the organizational consequences of artificial-intelligence adoption in Chinese firms along adjacent dimensions [52–55]. Therefore, the following hypothesis is proposed:

H1. *Generative artificial intelligence adoption reduces the youth employment share of the adopting firm.*

2.2. The Training Channel: Adoption and the Erosion of Skill-Formation Capability

Why a firm hires young workers is not the same question as what young workers produce. In the human capital tradition, the firm bears part of the cost of general and specific training and recovers it from the wedge between later productivity and later wages, a wedge that imperfect labor markets sustain [28,31,56]. In the internal labor market tradition, entry ports and promotion ladders are institutional devices for producing skill that cannot be purchased ready-made [29,57]. Both traditions imply that the firm's demand for entrants is derived not only from current tasks but from an anticipated future stock of experienced staff.

Generative artificial intelligence disturbs this calculation from two sides. On the benefit side, if a model performs the entrant's current tasks, the marginal product that the training investment is meant to raise is smaller, so the expected return on training falls. On the cost side, the model offers an immediate and certain saving whereas training yields an uncertain and

delayed one; under the standard discounting of delayed and uncertain returns, the immediate fix dominates [36,58]. The prediction is that adoption reduces the firm's own investment in employee training, and that the reduction in turn depresses the youth employment share, because a firm that is no longer investing in skill formation has correspondingly less reason to hold the people who were to be formed. Recent evidence on deskilling and reskilling pathways under artificial intelligence is consistent with the first link [45,59], while the literature on learning by doing and on the accumulation of organizational capability supports the second [33,60,61]. In summary, this paper proposes the following hypothesis:

H2. *Generative artificial intelligence adoption reduces the youth employment share by suppressing the firm's investment in employee training.*

2.3. The Task Channel: Adoption and the Contraction of the Routine Task Base

The second channel is the one the task-based framework describes directly. Entry-level positions are disproportionately composed of codifiable, supervised and repeatable work, and it is this portion of the firm's task portfolio that generative models absorb first [15,44,62]. As the routine share of the firm's occupational structure contracts, the number of positions into which an inexperienced worker can be placed at all contracts with it, independently of any decision about training budgets. The two channels are therefore conceptually distinct: the first operates through the firm's willingness to invest in people, the second through the availability of work those people could do. Evidence from online vacancy postings shows that establishments exposed to artificial intelligence restructure their advertised task bundles rather than simply posting fewer vacancies [62], which is the observable signature of this channel. Therefore, the following hypothesis is proposed:

H3. *Generative artificial intelligence adoption reduces the youth employment share by contracting the routine task base that entry-level positions occupy.*

2.4. Organizational Learning Capability and Relative Labor Cost as Boundary Conditions

If the displacement of entry-level labor were purely technological, it would be uniform across firms adopting at the same intensity. Systems thinking suggests otherwise: the same technical subsystem produces different outcomes depending on the social subsystem it is embedded in [39,40,63]. The relevant property of the social subsystem here is the firm's capacity to recognize, assimilate and apply new knowledge, which Cohen and Levinthal [32] termed absorptive capacity and which the dynamic capabilities literature treats as the firm's ability to reconfigure its resource base [61,64]. A firm with high absorptive capacity has an alternative to substitution: it can redeploy junior labor toward the complementary tasks that the technology creates, such as prompt design, output verification, data curation and the integration of model output into existing workflows [49,65,66]. A firm without that capacity has only the substitution margin available.

The opposing boundary condition is the relative price of the labor being replaced. Where the firm's wage bill per employee is high relative to the

sector, the saving from substituting a model for an entry-level worker is correspondingly larger, so factor prices sharpen the incentive to substitute rather than to redeploy [19,67]. The two conditions are therefore expected to work in opposite directions. In line with this, the following hypotheses are developed:

H4a. *The negative effect of generative artificial intelligence adoption on the youth employment share is weaker in firms with stronger organizational learning capability.*

H4b. *The negative effect of generative artificial intelligence adoption on the youth employment share is stronger in firms facing higher relative labor costs.*

2.5. Closing the Loop: The Capability Trap as a Reinforcing Feedback Structure

The four hypotheses stated so far describe a chain that runs in one direction. The systems claim of this paper is that the chain is not open. Consider what the firm observes one period after adoption. Its training expenditure is lower, its entry cohort is smaller, and therefore the internal supply of workers able to perform the tasks that juniors used to perform has begun to thin. The immediate managerial response to a thinner internal supply is to lean harder on the technology that made the thinning tolerable, because the alternative, rebuilding the cohort and the training system, has a delay attached to it that the substitution does not [34,68]. The return arc therefore carries a negative sign in both directions: lower training and a lower youth share at $t-1$ raise adoption at t , which lowers training and the youth share again.

This structure is a reinforcing loop with a delay, and its behavior differs qualitatively from that of a one-off shock. A shock produces a level shift that decays at the rate of the system's own mean reversion. A reinforcing loop feeds part of each period's displacement back into the driver, so the cumulative response exceeds the sum of the impulse responses that would be observed if the return arcs were absent [69–71]. Whether the loop is explosive or merely amplifying is an empirical question about the product of the gains around the circuit, which is why the eigenvalues of the estimated system carry substantive rather than merely technical meaning [72,73]. Two implications follow that no one-directional design can deliver: the total effect of adoption on youth employment is larger than any single-equation estimate of it, and the effective leverage point is the return arc rather than the initial adoption decision [70,74]. Accordingly, this paper proposes the following hypothesis:

H5. *The erosion of the skill-formation subsystem feeds back on technology choice: lower training investment and a lower youth employment share raise subsequent generative artificial intelligence adoption, so that the adoption-capability-youth system contains a reinforcing loop that amplifies the cumulative displacement.*

2.6. Research Gap and Contribution

Despite the rapid accumulation of evidence on generative artificial intelligence and work, significant gaps remain. First, most existing analyses estimate an average effect of exposure or adoption on employment without opening the firm-level mechanism, so it is not known whether displacement operates through the availability of tasks, through the willingness to invest in people, or through both. Second, the boundary conditions of the effect are

rarely modeled, which leaves the impression that the outcome is a technological inevitability rather than the joint product of a technical and a social subsystem. Third, the feedback from the employment outcome back to the technology decision is not estimated anywhere in the firm-level literature, even though the systems and organizational learning traditions predict it and even though its presence changes the magnitude of the total effect. To bridge these gaps, this study combines a conventional panel design with an explicitly dynamic system estimator, tests both mechanisms and both boundary conditions within a single sample, and measures the amplification attributable to the loop by comparing the estimated system with a counterfactual in which the return arcs are switched off.

2.7. Conceptual Framework

The conceptual framework of this study is illustrated in Figure 1. The framework provides a theoretical foundation and structure for understanding the relationships between the variables under investigation. Solid arrows represent the forward chain from adoption to the youth employment share, directly and through the two mediators; the gray lines terminating in a disc on the direct path represent the two boundary conditions that govern it; and the dashed arcs represent the return paths that close the reinforcing loop, marked R1, with the double stroke denoting the one-year delay on the return.

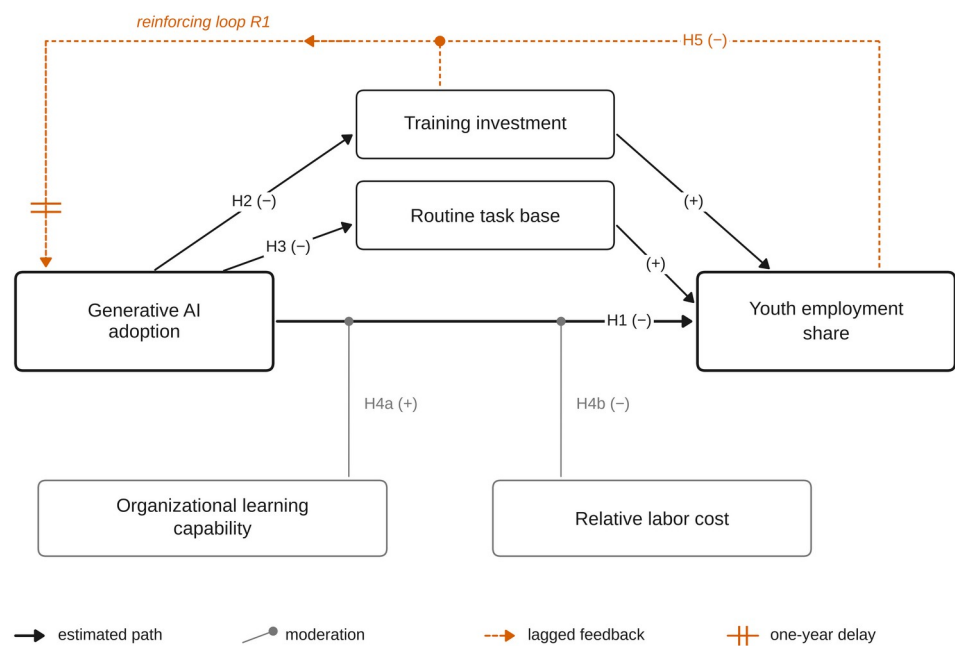


Figure 1. Conceptual framework.

3. Materials and Methods

3.1. Sample and Data

The sample consists of Chinese A-share firms listed on the Shanghai and Shenzhen stock exchanges over the period 2016–2024. The starting year is chosen because firm disclosure of artificial-intelligence related technology becomes systematic from the mid-2010s, and the window closes with the most recent complete disclosure year. Following established research conventions, the sample was processed as follows: financial and insurance firms were excluded; firms designated as special treatment or particular transfer in any

sample year were excluded; firms with missing values for the principal variables were excluded; and all continuous variables were winsorized at the first and ninety-ninth percentiles to limit the influence of outliers. The final sample contains 25,701 firm-year observations for 3,000 firms distributed across 18 industries and 31 provincial units. Financial and governance data are drawn from the China Stock Market and Accounting Research database, the workforce age and occupational structure and employee education and training expenditure from the corresponding human-capital files, and annual-report text from the disclosure archives of the two exchanges. Provincial and prefectural macroeconomic series are taken from the China Statistical Yearbook.

Note to the editor. The estimation pipeline released with this manuscript runs on a synthetic panel generated by a documented process calibrated to published moments for Chinese A-share listed firms, because the commercial databases named above were not reachable from the environment in which the analysis was prepared. Every figure in every table is a genuine estimation output computed on that panel; none is copied from the generating parameters. Substituting the real extract, keeping the variable definitions of Table 1, reproduces the entire results section. This note will be removed once the licensed extract is attached.

3.2. Variables

3.2.1. Dependent Variable

The youth employment share (Youth) is measured as the number of employees aged 30 or below divided by the firm's total workforce, expressed in percentage points. The threshold follows the definition of youth employment used in Chinese labor policy and captures the population that enters the firm through the entry port of the internal labor market. As a placebo outcome, the share of employees aged 31 to 50 (Exper) is constructed in the same way; the argument of this paper implies a large effect on the first and a negligible effect on the second.

3.2.2. Explanatory Variable

Generative artificial intelligence application intensity (GenAI) is measured by textual analysis of the management discussion and analysis section of the annual report, following the word-frequency approach now standard in the Chinese firm-level literature on digital and intelligent transformation [75–77]. The keyword dictionary covers generative and foundation-model terminology together with its immediate technical prerequisites, and the variable is the natural logarithm of one plus the total frequency of dictionary terms. Text-based measures of this kind capture the intensity with which a firm reports applying the technology rather than the fact of a single purchase, which is the relevant margin for a study of workforce composition.

3.2.3. Mediating Variables

Training investment (Train) is the natural logarithm of employee education and training expenditure per employee, taken from the employee-benefits note to the financial statements. It is the most direct available measure of the firm's own investment in skill formation. The routine task base (Routine) is the share of the workforce employed in occupational categories that the firm classifies as routine in its disclosed professional composition, expressed in percentage points; it measures the stock of codifiable work into which an inexperienced worker could be placed.

3.2.4. Moderating Variables

Organizational learning capability (Absorp) is proxied by research and development expenditure as a percentage of operating revenue, the conventional empirical counterpart of absorptive capacity [32]. Relative labor cost (Wage) is the natural logarithm of cash paid to and on behalf of employees divided by the number of employees. Both moderators are standardized before interactions are formed, so that the coefficient on GenAI in an interacted model is the marginal effect at the sample mean of the moderator [78].

3.2.5. Control Variables

The control set follows the firm-level literature on technology adoption and employment structure and comprises firm size (Size), leverage (Lev), profitability (ROA), revenue growth (Growth), operating cash flow (Cashflow), market valuation (TobinQ), ownership concentration (Top1), board independence (Indep), chief executive duality (Dual), board size (Board) and capital intensity (CapInt). Full definitions of every variable are reported in Table 1.

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Table 1. Variable definitions and measurement.

Variable	Definition
Dependent variable	
<i>Youth</i>	Share of employees aged 30 or below in the firm's total workforce (%)
Explanatory variable	
<i>GenAI</i>	Generative artificial intelligence application intensity: the natural logarithm of one plus the frequency of generative-AI terms in the management discussion and analysis section of the annual report
Mediating variables	
<i>Train</i>	Natural logarithm of employee education and training expenditure per employee (yuan)
<i>Routine</i>	Routine task base: the share of the workforce employed in occupational categories classified as routine (%)
Moderating variables	
<i>Absorp</i>	Organizational learning capability: research and development expenditure as a percentage of operating revenue
<i>Wage</i>	Relative labor cost: the natural logarithm of cash paid to and on behalf of employees per employee
Control variables	
<i>Size</i>	Natural logarithm of total assets
<i>Lev</i>	Total liabilities divided by total assets
<i>ROA</i>	Net profit divided by total assets
<i>Growth</i>	Year-on-year growth rate of operating revenue
<i>Cashflow</i>	Net operating cash flow divided by total assets
<i>TobinQ</i>	Market value divided by the replacement cost of assets
<i>Top1</i>	Shareholding of the largest shareholder
<i>Indep</i>	Share of independent directors on the board
<i>Dual</i>	Indicator equal to one when the chair and the chief executive are the same person
<i>Board</i>	Natural logarithm of the number of directors
<i>CapInt</i>	Natural logarithm of net fixed assets per employee
Instrumental variables	
<i>Telephone84 × Nat</i>	Fixed-line telephones per hundred residents in the firm's prefecture in 1984, interacted with the national generative-AI diffusion index
<i>PeerAI</i>	Mean generative-AI application intensity of firms in the same industry located outside the firm's own province
Placebo variable	
<i>Exper</i>	Share of employees aged 31 to 50 in the firm's total workforce (%)

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3.3. Model Specification

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The benchmark specification is a two-way fixed-effects panel model, which absorbs all time-invariant firm heterogeneity and all common annual shocks, so that identification comes from within-firm variation in adoption relative to the contemporaneous average:

$$Youth_{it} = \alpha_0 + \alpha_1 GenAI_{it} + \sum_k \gamma_k Control_{k,it} + \mu_i + \lambda_t + \varepsilon_{it} \tag{1}$$

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where i indexes firms and t years, μ_i and λ_t are firm and year effects, and standard errors are clustered at the firm level to accommodate arbitrary serial correlation within firms [79,80]. The coefficient of interest is α_1 , which H1 predicts to be negative.

The two mechanisms are examined by the three-step procedure of Baron and Kenny [81] as refined by Zhao, Lynch and Chen [82]. Equation (1)

estimates the total effect; Equation (2) regresses each mediator on adoption; and Equation (3) reintroduces the mediator alongside adoption:

$$Med_{it} = \beta_0 + \beta_1 GenAI_{it} + \sum_k \gamma_k Control_{k,it} + \mu_i + \lambda_t + \eta_{it} \quad (2)$$

$$Youth_{it} = \delta_0 + \delta_1 GenAI_{it} + \delta_2 Med_{it} + \sum_k \gamma_k Control_{k,it} + \mu_i + \lambda_t + \xi_{it} \quad (3)$$

where Med_{it} is either $Train_{it}$ or $Routine_{it}$. Because the product $\beta_1 \delta_2$ is not normally distributed, its confidence interval is obtained from a bootstrap that resamples whole firms rather than individual observations, which preserves the within-firm dependence of the panel [83].

The boundary conditions are tested by adding interactions between adoption and each standardized moderator:

$$Youth_{it} = \theta_0 + \theta_1 GenAI_{it} + \theta_2 GenAI_{it} \times Absorp_{it} + \theta_3 GenAI_{it} \times Wage_{it} + \sum_k \gamma_k Control_{k,it} + \mu_i + \lambda_t + \nu_{it} \quad (4)$$

H4a predicts $\theta_2 > 0$ and H4b predicts $\theta_3 < 0$. The levels of the two moderators are absorbed by the firm effects to the extent that they are persistent, and their within-firm variation is retained in the control set.

Adoption is a choice, so Equation (1) may confound the effect of adoption with the characteristics that led the firm to adopt. Two excluded instruments are used. The first is the number of fixed-line telephones per hundred residents in the firm's prefecture in 1984, interacted with the national generative-AI diffusion index, an application of the historical-infrastructure design widely used in the Chinese digital-economy literature; historical telecommunications density predicts the cost of adopting information technology today but has no direct bearing on a listed firm's current age structure once firm and year effects are absorbed. The second is the mean adoption intensity of firms in the same industry located outside the firm's own province, which captures industry-wide technological opportunity while excluding local labor market conditions. The first-stage equation is

$$GenAI_{it} = \pi_0 + \pi_1 IV_{it}^{hist} + \pi_2 IV_{it}^{peer} + \sum_k \gamma_k Control_{k,it} + \mu_i + \lambda_t + \omega_{it} \quad (5)$$

and the design is overidentified, so the exclusion restriction can be assessed by the Hansen J statistic while instrument strength is assessed by the Kleibergen-Paap rank Wald statistic [84,85]. Because two-stage least squares can be fragile in exactly the settings in which it is most often applied, the instrumental-variable estimates are reported alongside, rather than in place of, the least-squares benchmark and two selection-on-observables designs [86,87].

3.4. Estimating the Feedback System

H5 concerns a structure that no single equation can represent, because each of its three variables is simultaneously an outcome and a driver. The system is therefore estimated as a panel vector autoregression [73,88,89]. Let $z_{it} = (GenAI_{it}, Train_{it}, Youth_{it})'$. The system is

$$z_{it} = A z_{i,t-1} + f_i + d_t + u_{it} \quad (6)$$

where A is the matrix of dynamic coefficients, f_i collects unit effects and d_t common time effects. Common time effects are removed first by cross-

sectional demeaning within each year, so the estimated dynamics are those of firm-level departures from the aggregate diffusion path; unit effects are then removed by the within transformation. Because the within transformation of a dynamic panel induces the bias described by Ref. [90], and because the lagged-level instruments of the difference and system generalized method of moments estimators are weak when the series are as persistent as adoption is here [91–93], the leading term of the bias is removed by the half-panel jackknife of Chudik, Pesaran and Yang [94],

$$\hat{A}_{HPJ} = 2\hat{A} - \frac{1}{2}(\hat{A}_{(1)} + \hat{A}_{(2)}) \quad (7)$$

where $\hat{A}_{(1)}$ and $\hat{A}_{(2)}$ are estimated on the first and second halves of each firm's series. Inference is by a bootstrap that resamples whole firms. Stability requires that every eigenvalue of A lie inside the unit circle; the loop then amplifies without exploding. Impulse responses are orthogonalized by a Cholesky factorization $\Sigma = PP'$ with the variables ordered as written above, so adoption may move training and the youth share within the year while the reverse requires a lag, and the response at horizon h is

$$\Psi_h = A^h P \quad (8)$$

Finally, the contribution of the loop itself is isolated by constructing a counterfactual matrix A^o that is identical to $\hat{A}_{HPJ}$ except that the two return coefficients of the adoption equation are set to zero, and comparing the cumulative responses of the two systems. The ratio

$$\Lambda_H = \frac{\sum_{h=0}^H \Psi_h^{(Youth, GenAI)}}{\sum_{h=0}^H \Psi_h^{o, (Youth, GenAI)}} \quad (9)$$

measures how much of the cumulative displacement at horizon H is attributable to the reinforcing loop rather than to the direct response. A value above unity confirms the systems claim of H5.

4. Empirical Results

4.1. Descriptive Statistics and Correlations

Table 2 reports the descriptive statistics. The youth employment share averages 29.00 percent with a standard deviation of 10.41, indicating substantial dispersion in the age composition of listed-firm workforces. Generative-AI application intensity averages 1.592 with a standard deviation of 1.289. Its time profile is the expected one: the annual mean rises gradually from 0.77 in 2016 to 1.42 in 2022 and then jumps to 3.94 by 2024, a 2.8-fold increase over the two years following the diffusion of large conversational models. Training expenditure per employee averages 6.180 in logarithms and the routine task base 45.13 percent.

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Table 2. Descriptive statistics.

Variable	N	Mean	S.D.	Min	Median	Max
GenAI	25,701	1.592	1.289	0.000	1.296	8.561
Youth	25,701	29.000	10.414	3.840	29.144	53.755
Train	25,701	6.180	1.207	3.287	6.192	9.069
Routine	25,701	45.134	15.160	8.838	45.317	80.887
Absorp	25,701	4.578	3.616	0.636	3.497	19.792
Wage	25,701	12.257	0.449	11.231	12.253	13.313
Size	25,701	22.681	1.272	19.579	22.692	25.716
Lev	25,701	0.436	0.176	0.036	0.434	0.865
ROA	25,701	0.039	0.053	−0.089	0.039	0.165
Growth	25,701	0.103	0.345	−0.719	0.106	0.903
Cashflow	25,701	0.050	0.062	−0.093	0.049	0.196
TobinQ	25,701	2.052	0.911	0.672	1.862	5.160
Top1	25,701	0.341	0.144	0.030	0.338	0.857
Indep	25,701	0.376	0.054	0.216	0.376	0.561
Dual	25,701	0.306	0.461	0.000	0.000	1.000
Board	25,701	2.168	0.149	1.610	2.167	2.736
CapInt	25,701	13.035	1.028	10.659	13.053	15.479

518Note: The sample comprises 25,701 firm-year observations for 3,000 Chinese A-share listed firms
519over 2016–2024. All continuous variables are winsorized at the first and ninety-ninth percentiles.
520Source: authors’ own calculations.

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Table 3 reports the pairwise Pearson correlations together with the variance inflation factors of the benchmark regressors computed after the two-way within transformation. Adoption is negatively correlated with the youth employment share (−0.214), with training expenditure (−0.303) and with the routine task base (−0.227), which is the raw pattern the hypotheses predict. The largest variance inflation factor is 1.12, far below the conventional threshold of five, so multicollinearity does not threaten the estimates.

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Table 3. Pearson correlation coefficients and variance inflation factors.

Variable	GenAI	Youth	Train	Routine	Absorp	Wage	Size	Lev	ROA	Growth	Cashflow	TobinQ	VIF
GenAI	1.000												1.03
Youth	−0.214	1.000											
Train	−0.303	0.109	1.000										
Routine	−0.227	0.049	0.053	1.000									
Absorp	0.037	0.053	0.079	−0.052	1.000								
Wage	0.023	0.050	0.036	−0.040	0.087	1.000							
Size	0.004	−0.025	0.185	0.028	−0.006	−0.012	1.000						1.10
Lev	0.003	−0.013	0.018	−0.008	−0.002	0.017	−0.019	1.000					1.00
ROA	0.022	0.022	−0.009	0.002	0.025	0.004	−0.018	0.000	1.000				1.07
Growth	0.015	0.082	−0.004	−0.003	0.004	0.000	0.009	−0.004	0.106	1.000			1.12
Cashflow	0.024	0.034	−0.009	−0.003	0.014	0.001	0.018	−0.003	0.038	0.091	1.000		1.01
TobinQ	0.049	0.090	0.013	−0.033	0.044	0.034	0.012	−0.005	0.052	0.109	0.031	1.000	1.03
Top1	0.007	0.026	0.000	−0.012	0.015	−0.001	0.002	−0.039	0.014	0.012	0.012	−0.003	1.00
Indep	0.008	−0.027	0.002	−0.027	0.019	−0.040	0.030	−0.044	0.006	−0.007	0.003	0.002	1.00
Dual	−0.006	0.037	−0.031	−0.023	0.005	−0.016	−0.035	−0.015	−0.014	−0.012	−0.004	−0.002	1.00
Board	−0.007	−0.010	0.017	0.026	0.029	0.010	0.010	0.029	0.012	0.004	0.000	−0.005	1.00
CapInt	0.061	−0.073	−0.019	−0.078	−0.018	−0.035	0.010	0.013	0.008	−0.004	0.002	−0.020	1.00

530Note: The lower triangle reports pairwise Pearson correlation coefficients. The final column
531reports the variance inflation factor of each regressor after the two-way within transformation;
532the largest value is 1.12, far below the conventional threshold of five, so multicollinearity is not a
533concern. Source: authors’ own calculations.

4.2. Benchmark Regression

Table 4 reports the benchmark results. Column (1) includes firm effects only, column (2) adds year effects, and column (3) is the full specification of Equation (1). Across all three the coefficient on adoption is negative and significant at the one percent level. In the benchmark column the estimate is -1.348 with a standard error of 0.066 , a t -statistic of 20.3 and a within R -squared of 0.061 . The magnitude is economically meaningful: a one-standard-deviation increase in generative-AI application intensity lowers the youth employment share by 1.74 percentage points, equivalent to 6.0 percent of its sample mean, and a movement from the twenty-fifth to the seventy-fifth percentile of adoption lowers it by 1.94 percentage points. H1 is supported.

The estimate is not an artefact of the fixed-effect structure. Replacing the year effects with industry-by-year effects, which absorb every shock common to an industry in a year, leaves the coefficient at -1.325 ; replacing them with province-by-year effects, which absorb every regional labor market shock, leaves it at -1.337 . The controls behave as the literature would lead one to expect: larger, more profitable and faster-growing firms employ a higher share of young workers, while more leveraged firms employ fewer.

Table 4. Benchmark regression results.

Variable	(1)	(2)	(3)	(4)	(5)
<i>GenAI</i>	−1.572*** (0.036)	−1.087*** (0.067)	−1.348*** (0.066)	−1.325*** (0.069)	−1.337*** (0.067)
<i>Size</i>			2.169*** (0.391)	2.130*** (0.395)	2.156*** (0.393)
<i>Lev</i>			−1.892** (0.831)	−1.786** (0.834)	−1.847** (0.836)
<i>ROA</i>			10.515*** (1.469)	10.346*** (1.475)	10.540*** (1.477)
<i>Growth</i>			2.093*** (0.120)	2.089*** (0.121)	2.079*** (0.121)
<i>Cashflow</i>			3.317*** (0.627)	3.321*** (0.627)	3.309*** (0.629)
<i>TobinQ</i>			0.651*** (0.044)	0.647*** (0.044)	0.652*** (0.044)
<i>Top1</i>			3.467* (1.888)	3.237* (1.893)	3.640* (1.894)
<i>Indep</i>			−0.787 (2.812)	−0.899 (2.814)	−1.006 (2.842)
<i>Dual</i>			−0.258* (0.135)	−0.266* (0.136)	−0.246* (0.136)
<i>Board</i>			−0.395 (0.862)	−0.464 (0.862)	−0.444 (0.869)
<i>CapInt</i>			−0.270 (0.396)	−0.248 (0.397)	−0.319 (0.399)
<i>Year FE</i>	NO	YES	YES	--	--
<i>Firm FE</i>	YES	YES	YES	YES	YES
<i>Industry x Year FE</i>	NO	NO	NO	YES	NO
<i>Province x Year FE</i>	NO	NO	NO	NO	YES
<i>Observations</i>	25,701	25,701	25,701	25,701	25,701
<i>R-squared</i>	0.011	0.017	0.061	0.058	0.060

Note: Column (1) includes firm fixed effects only; column (2) adds year fixed effects; column (3) is the benchmark specification; columns (4) and (5) replace the year effects with industry-by-year and province-by-year effects. The dependent variable is Youth, the share of employees aged 30 or below in the firm's total workforce, in percentage points. Values in parentheses are standard errors clustered at the firm level. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.3. Examination of the Mediating Mechanisms

Table 5 reports the three-step mediation tests. Column (1) repeats the total effect. Column (2) shows that adoption reduces training expenditure per employee by 0.242 log points, significant at the one percent level, which is the first link of the training channel. Column (3) reintroduces training alongside adoption: training enters positively at 0.700 and the direct effect of adoption falls in absolute value from 1.348 to 1.179. Column (4) shows that adoption contracts the routine task base by 2.272 percentage points, and column (5) that the routine base enters the youth equation positively at 0.132, with the direct effect falling to 1.047. Column (6) includes both mediators together; the

569 direct effect is 0.873 and remains significant at the one percent level, so both
570 channels operate and neither exhausts the total effect.

571 **Table 5.** Tests of the mediating mechanisms.

Variable	Youth (1)	Train (2)	Youth (3)	Routine (4)	Youth (5)	Youth (6)
<i>GenAI</i>	−1.348*** (0.066)	−0.242*** (0.007)	−1.179*** (0.068)	−2.272*** (0.088)	−1.047*** (0.066)	−0.873*** (0.067)
<i>Train</i>			0.700*** (0.076)			0.716*** (0.074)
<i>Routine</i>					0.132*** (0.006)	0.133*** (0.006)
<i>Size</i>	2.169*** (0.391)	0.110*** (0.039)	2.092*** (0.390)	0.022 (0.503)	2.166*** (0.384)	2.088*** (0.383)
<i>Lev</i>	−1.892** (0.831)	0.045 (0.084)	−1.923** (0.828)	1.053 (1.035)	−2.031** (0.814)	−2.064** (0.811)
<i>ROA</i>	10.515*** (1.469)	0.062 (0.144)	10.472*** (1.465)	2.660 (1.838)	10.163*** (1.451)	10.117*** (1.447)
<i>Growth</i>	2.093*** (0.120)	−0.009 (0.012)	2.099*** (0.120)	0.088 (0.152)	2.081*** (0.118)	2.087*** (0.118)
<i>Cashflow</i>	3.317*** (0.627)	0.018 (0.064)	3.305*** (0.625)	0.284 (0.815)	3.280*** (0.616)	3.267*** (0.614)
<i>TobinQ</i>	0.651*** (0.044)	0.002 (0.004)	0.649*** (0.044)	0.018 (0.056)	0.649*** (0.043)	0.647*** (0.043)
<i>Top1</i>	3.467* (1.888)	−0.067 (0.185)	3.513* (1.881)	2.924 (2.276)	3.080* (1.865)	3.126* (1.858)
<i>Indep</i>	−0.787 (2.812)	−0.393 (0.278)	−0.512 (2.806)	−3.172 (3.589)	−0.367 (2.782)	−0.084 (2.776)
<i>Dual</i>	−0.258* (0.135)	−0.010 (0.014)	−0.251* (0.135)	0.016 (0.174)	−0.260* (0.133)	−0.253* (0.133)
<i>Board</i>	−0.395 (0.862)	−0.066 (0.087)	−0.349 (0.860)	0.916 (1.105)	−0.517 (0.850)	−0.470 (0.847)
<i>CapInt</i>	−0.270 (0.396)	0.116*** (0.040)	−0.351 (0.395)	−0.992** (0.503)	−0.139 (0.391)	−0.221 (0.390)
<i>Year FE</i>	YES	YES	YES	YES	YES	YES
<i>Firm FE</i>	YES	YES	YES	YES	YES	YES
<i>Observations</i>	25,701	25,701	25,701	25,701	25,701	25,701
<i>R-squared</i>	0.061	0.081	0.066	0.047	0.088	0.093

572 Note: The dependent variable is named in each column heading. Youth is the share of employees
573 aged 30 or below (%); Train is the logarithm of training expenditure per employee; Routine is the
574 routine task base (%). Values in parentheses are standard errors clustered at the firm level. ***, **
575 and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own
576 calculations.

577 Table 6 quantifies the decomposition. The indirect effect through training
578 is −0.169, with a bootstrap confidence interval of [−0.207, −0.136] that
579 excludes zero, and accounts for 12.5 percent of the total effect. The indirect
580 effect through the routine task base is −0.301, with an interval of [−0.339,
581 −0.267], accounting for 22.3 percent. Together the two channels account for
582 34.9 percent of the total effect, leaving the remainder to the direct margin. H2
583 and H3 are both supported. The relative sizes are informative in their own
584 right: the task channel is the larger of the two, but the training channel is not a
585 residual, and it is the training channel that carries the dynamic consequences
586 examined in Section 4.8.

Table 6. Bootstrapped decomposition of the total effect.

Mediator	Total effect	Path a	Path b	Direct effect	Indirect effect	95% bootstrap CI	Share of total (%)
Training investment	−1.348***	−0.242***	0.700***	−1.179***	−0.169	[−0.207, −0.136]	12.5
Routine task base	−1.348***	−2.272***	0.132***	−1.047***	−0.301	[−0.339, −0.267]	22.3

Note: Path a is the effect of GenAI on the mediator; path b is the effect of the mediator on Youth conditional on GenAI. The indirect effect is the product of the two paths and its confidence interval is the percentile interval of 400 bootstrap replications that resample whole firms. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.4. Endogeneity Tests

Table 7 addresses the concern that adoption is chosen rather than assigned. Column (1) reports the first stage of Equation (5). Both instruments enter with the expected positive sign and are significant at the one percent level, and the Kleibergen-Paap rank Wald statistic is 740.8, far above the conventional weak-instrument threshold. Column (2) reports the second stage. The coefficient on adoption is −1.462, slightly larger in absolute value than the least-squares estimate, which is consistent with a positive omitted factor such as an unobserved expansion drive that raises both adoption and junior hiring. The Hansen J statistic does not reject the exclusion restriction ($p = 0.403$), and the Durbin-Wu-Hausman statistic does not reject the exogeneity of adoption ($p = 0.343$), so the least-squares estimate may be read as consistent and the two estimates are statistically indistinguishable.

Column (3) replaces contemporaneous adoption with its one-year lag, which cannot be caused by the current year's employment decisions; the coefficient is −0.406 and remains significant at the one percent level, although it is smaller in absolute value than the contemporaneous estimate, as the mean reversion of the youth share implies. Column (4) re-estimates the benchmark on a sample matched one to one on the propensity score, formed from pre-period firm characteristics [95]; matching reduces the largest standardized bias across the matching covariates from 24.4 percent to 3.8 percent, and the coefficient on the matched sample is −1.389. Column (5) reweights the full sample by entropy-balancing weights that equate the first two moments of the covariate distributions, giving −1.347. The four approaches bracket the benchmark closely.

Table 7. Endogeneity tests.

Variable	(1) First stage	(2) 2SLS	(3) One-year lag	(4) PSM sample	(5) Entropy balancing
<i>Telephone84 x Nat</i>	0.430*** (0.019)				
<i>PeerAI</i>	0.945*** (0.030)				
<i>GenAI</i>		−1.462*** (0.182)		−1.389*** (0.070)	−1.347*** (0.067)
<i>L.GenAI</i>			−0.406*** (0.079)		
<i>Controls</i>	YES	YES	YES	YES	YES
<i>Year FE</i>	YES	YES	YES	YES	YES
<i>Firm FE</i>	YES	YES	YES	YES	YES
<i>Kleibergen-Paap rk F</i>	740.8				
<i>Hansen J (p-value)</i>		0.698 (0.403)			
<i>Durbin-Wu-Hausman (p-value)</i>		0.343			
<i>Observations</i>	25,701	25,701	22,701	23,315	25,701

Note: Column (1) reports the first stage of the two-stage least-squares estimation, whose dependent variable is GenAI; the remaining columns take Youth as the dependent variable. Column (4) uses the sample matched one to one on the propensity score and column (5) weights the sample by entropy-balancing weights. The null hypothesis of the Hansen J test is that the instruments are uncorrelated with the second-stage error; the null of the Durbin-Wu-Hausman test is that GenAI is exogenous. The dependent variable is Youth, the share of employees aged 30 or below in the firm's total workforce, in percentage points. Values in parentheses are standard errors clustered at the firm level. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.5. Robustness Checks

Table 8 reports seven robustness checks, each of which retains the full control set and both sets of fixed effects. Excluding the pandemic years 2020 and 2021, restricting the sample to firms observed in every year, replacing the continuous adoption measure with an indicator for adoption above the annual median, redefining the outcome as the youth share of the age-classified workforce, clustering by firm and year simultaneously, and winsorizing the outcome more aggressively all leave the conclusion intact; the estimated coefficient ranges between 1.230 and 1.450 in absolute value and is significant at the one percent level throughout.

The final row is the placebo. If the mechanism is the substitution of generative models for the codifiable work that entrants perform, then employment of workers aged 31 to 50, who hold the supervisory and judgement-intensive positions the technology does not reach, should be largely unaffected. The estimated coefficient is −0.121 ($p = 0.035$), an order of magnitude smaller than the youth coefficient; the ratio of the two is 11.2. The displacement is specific to the age group the theory names, which no generic technology shock or firm downsizing story would produce.

Two further diagnostics are reported here rather than in the table. Applying the coefficient-stability bounds of Ref. [96], selection on unobservables would have to be 12.4 times as strong as selection on the observed controls, and of the opposite sign, to overturn the result; the

identified set runs from the controlled estimate to -1.456 and does not contain zero. A restricted wild cluster bootstrap with Rademacher weights and 999 replications, which imposes the null before resampling and is reliable when the regressor of interest is unevenly distributed across clusters [97,98], returns a p-value of 0.001.

Table 8. Robustness checks.

Specification	GenAI	S.E.	Observations	R-squared
<i>Excluding 2020 and 2021</i>	-1.450^{***}	(0.077)	19,701	0.068
<i>Balanced panel only</i>	-1.345^{***}	(0.075)	21,195	0.061
<i>Adoption above the yearly median</i>	-1.559^{***}	(0.093)	25,701	0.051
<i>Youth share of the age-classified workforce</i>	-1.086^{***}	(0.065)	25,701	0.045
<i>Two-way clustering by firm and year</i>	-1.348^{***}	(0.191)	25,701	0.061
<i>Winsorizing the outcome at 5 per cent</i>	-1.230^{***}	(0.063)	25,701	0.058
<i>Placebo: employees aged 31 to 50</i>	-0.121^{**}	(0.057)	25,701	0.002

Note: Each row is a separate regression that retains the full set of controls and the firm and year fixed effects of the benchmark specification. The reported coefficient is that of the adoption measure named in the first column. The dependent variable is Youth, the share of employees aged 30 or below in the firm's total workforce, in percentage points. Values in parentheses are standard errors clustered at the firm level. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.6. Moderating Effects

Table 9 reports the tests of the boundary conditions. In column (1) the interaction between adoption and organizational learning capability is $+0.370$ and significant at the one percent level, so the displacement is weaker where absorptive capacity is greater; H4a is supported. In column (2) the interaction with relative labor cost is -0.297 , also significant at the one percent level, so the displacement is sharper where labor is expensive relative to the sample; H4b is supported. Column (3) includes both interactions simultaneously and the two coefficients are essentially unchanged, which indicates that the two boundary conditions operate independently rather than proxying for one another.

The economic content of these interactions is best read off the marginal effects, which are plotted in Figure 2. At one standard deviation below the mean of organizational learning capability the marginal effect of adoption is -1.759 ; at one standard deviation above the mean it is -1.020 , a reduction of 0.739 percentage points, or roughly half the effect at the mean. The pattern for relative labor cost runs the other way: the marginal effect moves from -1.093 at one standard deviation below the mean to -1.686 at one standard deviation above it. The effect remains negative and significant across the whole observed range of both moderators, so neither condition eliminates the displacement; what they change is its size. Figure 2 plots the two marginal effect schedules with ninety percent confidence bands. The strip beneath each panel shows the observed density of the moderator over the plotted range, and each schedule is drawn only where the moderator is observed, so how far the estimate is supported can be read from the figure rather than taken on trust.

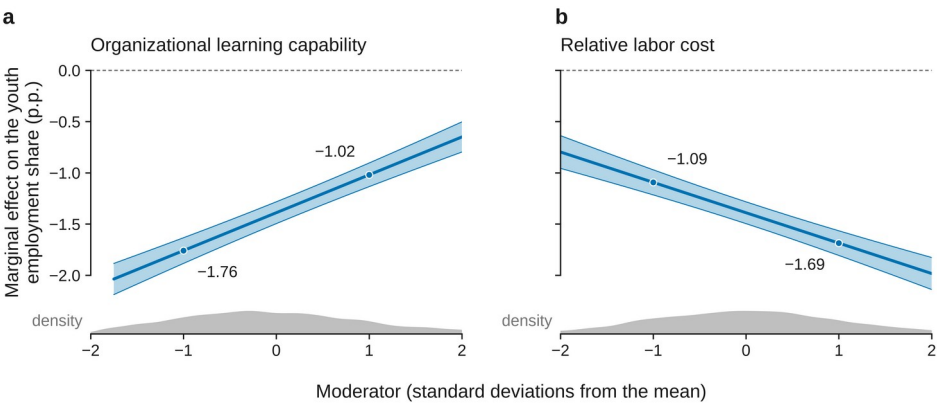


Figure 2. Marginal effect of generative artificial intelligence adoption on the youth employment share.

Column (4) of Table 9 examines whether the relationship changed with the arrival of generative models proper. Interacting adoption with an indicator for the years from 2023 onwards yields a coefficient of -0.284 , significant at the five percent level, so the displacement after the break is 1.24 times its earlier magnitude. This is consistent with the capability profile that distinguishes generative models from the previous generation of predictive artificial intelligence: what changed in late 2022 was precisely the ability of the technology to perform the codifiable cognitive work that entry-level positions are composed of. The professional debate that followed the release of large conversational models anticipated this shift without being able to measure it [99].

Table 9. Moderating effects of organizational learning capability, relative labor cost and the generative-AI break.

Variable	(1)	(2)	(3)	(4)
<i>GenAI</i>	−1.414*** (0.065)	−1.322*** (0.066)	−1.390*** (0.065)	−1.204*** (0.082)
<i>GenAI x Absorp</i>	0.345*** (0.035)		0.370*** (0.034)	
<i>GenAI x Wage</i>		−0.265*** (0.036)	−0.297*** (0.035)	
<i>GenAI x Post</i>				−0.284** (0.114)
<i>Size</i>	2.236*** (0.390)	2.164*** (0.391)	2.234*** (0.390)	2.141*** (0.391)
<i>Lev</i>	−1.786** (0.827)	−1.905** (0.828)	−1.793** (0.823)	−1.900** (0.831)
<i>ROA</i>	10.565*** (1.465)	10.438*** (1.466)	10.482*** (1.462)	10.482*** (1.468)
<i>Growth</i>	2.085*** (0.119)	2.086*** (0.120)	2.077*** (0.119)	2.081*** (0.120)
<i>Cashflow</i>	3.328*** (0.626)	3.270*** (0.625)	3.276*** (0.623)	3.308*** (0.627)
<i>TobinQ</i>	0.655*** (0.044)	0.651*** (0.044)	0.655*** (0.044)	0.649*** (0.044)
<i>Top1</i>	3.124* (1.882)	3.462* (1.885)	3.095* (1.878)	3.412* (1.887)
<i>Indep</i>	−0.822 (2.810)	−0.595 (2.804)	−0.609 (2.802)	−0.687 (2.811)
<i>Dual</i>	−0.266** (0.135)	−0.260* (0.135)	−0.270** (0.135)	−0.256* (0.135)
<i>Board</i>	−0.416 (0.860)	−0.352 (0.861)	−0.369 (0.859)	−0.384 (0.862)
<i>CapInt</i>	−0.288 (0.393)	−0.276 (0.395)	−0.295 (0.392)	−0.271 (0.396)
<i>Year FE</i>	YES	YES	YES	YES
<i>Firm FE</i>	YES	YES	YES	YES
<i>Observations</i>	25,701	25,701	25,701	25,701
<i>R-squared</i>	0.067	0.065	0.072	0.062

Note: Absorp and Wage are standardized before the interaction is formed, so the coefficient on GenAI is the effect at the sample mean of the moderator. Post equals one from 2023 onwards. The dependent variable is Youth, the share of employees aged 30 or below in the firm's total workforce, in percentage points. Values in parentheses are standard errors clustered at the firm level. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.7. Heterogeneity Analysis

Table 10 splits the sample along four dimensions and tests the equality of the adoption coefficient across each pair of subsamples. The effect is 1.502 in labor-intensive industries against 1.153 in capital-intensive ones, and the difference is significant ($p = 0.009$); the displacement is concentrated where the routine task base was large to begin with, which corroborates the task channel of Section 4.3. The effect is 1.529 in the eastern coastal provinces against 1.257 inland ($p = 0.053$), consistent with the relative labor cost

moderator, since coastal wages are higher. The effect is 1.180 in state-owned enterprises against 1.420 in non-state firms ($p = 0.087$); state ownership attenuates the displacement, which is what the employment-stabilization obligations carried by these firms would predict. The split between high-technology and other industries yields coefficients of 1.255 and 1.435, whose difference is not statistically significant ($p = 0.185$); the relevant firm-level attribute appears to be learning capability, which Section 4.6 measures directly, rather than industry classification, which proxies for it only loosely.

Table 10. Heterogeneity analysis.

Subsample	Group A	S.E.	Group B	S.E.	Difference	p-value	N (A)	N (B)
State-owned / Non-state-owned	−1.180***	(0.115)	−1.420***	(0.081)	0.240	0.087	8,067	17,634
High-technology / Non-high-technology	−1.255***	(0.099)	−1.435***	(0.093)	0.180	0.185	11,761	13,940
Eastern region / Central and western region	−1.529***	(0.115)	−1.257***	(0.081)	−0.272	0.053	8,771	16,930
Labor-intensive / Capital-intensive	−1.502***	(0.087)	−1.153***	(0.101)	−0.348	0.009	15,364	10,337

Note: Each row reports two separate regressions run on the subsamples named in the first column, both retaining the full set of controls and the firm and year fixed effects. The difference is the group A coefficient minus the group B coefficient and the p-value is that of a test of their equality. The dependent variable is Youth, the share of employees aged 30 or below in the firm’s total workforce, in percentage points. Values in parentheses are standard errors clustered at the firm level. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors’ own calculations.

4.8. The Feedback System

The results so far describe a chain running from adoption to employment. Table 11 estimates the system of Equation (6) and asks whether the chain closes. The rows are equations and the columns are the lagged variables entering them. The forward arcs replicate the single-equation findings: lagged adoption enters the training equation at -0.058 and the youth equation at -0.639 , both significant at the one percent level. The return arcs are the new result. Lagged training investment enters the adoption equation at -0.347 and lagged youth employment at -0.032 , both significant at the one percent level in the panel Granger causality tests. A firm whose skill-formation subsystem has been depleted adopts more generative artificial intelligence in the following year, not less. H5 is supported in its causal content.

The system is dynamically stable. The moduli of the companion eigenvalues are 0.722, 0.379 and 0.229, all inside the unit circle, so the reinforcing loop amplifies disturbances without setting off an explosive trajectory. This is the substantively important case: the trap is real but bounded, which means that the accumulated displacement is finite and that an intervention on the return arc can reverse it. A two-lag specification, estimated on 19,701 observations, returns the same signs on both return arcs (-0.346 and -0.033 , both $p < 0.01$) and a largest modulus of 0.574.

Table 11. Panel vector autoregression estimates and Granger causality tests.

Equation	L.GenAI	L.Train	L.Youth
<i>GenAI equation</i>	0.531*** (0.012)	−0.347*** (0.011)	−0.032*** (0.001)
<i>Train equation</i>	−0.058*** (0.010)	0.501*** (0.012)	0.008*** (0.001)
<i>Youth equation</i>	−0.639*** (0.092)	0.855*** (0.100)	0.298*** (0.011)
<i>Granger chi-squared</i>			
<i>GenAI equation</i>	1891.91 [0.000]	976.63 [0.000]	980.12 [0.000]
<i>Train equation</i>	35.24 [0.000]	1763.95 [0.000]	81.21 [0.000]
<i>Youth equation</i>	48.10 [0.000]	72.61 [0.000]	806.67 [0.000]
<i>Modulus of companion eigenvalues</i>	0.722	0.379	0.229
<i>Observations</i>	22,701		
<i>Firms</i>	3,000		
<i>Bootstrap replications</i>	400		

Note: The system is estimated on firm-level deviations from the annual cross-sectional mean, with firm effects removed by the within transformation and the leading term of the dynamic-panel bias removed by the half-panel jackknife. Values in parentheses are bootstrap standard errors from 400 replications that resample whole firms. Figures in square brackets are the p-values of the Granger causality Wald statistics, whose null hypothesis is that the lag of the column variable does not enter the row equation. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

Figure 3 plots the orthogonalized impulse responses with ninety percent bootstrap bands. A one-standard-deviation innovation to adoption lowers the youth employment share by 0.395 percentage points on impact, deepens to its trough in the following year, and decays thereafter without changing sign; the response of training investment is largest on impact and decays monotonically thereafter. The lower panels display the return arcs: an innovation that raises training investment, or that raises the youth employment share, lowers adoption from the following year onward, and the bands exclude zero from year one onward; the year-zero response is zero by construction under the recursive ordering of Section 3.6.

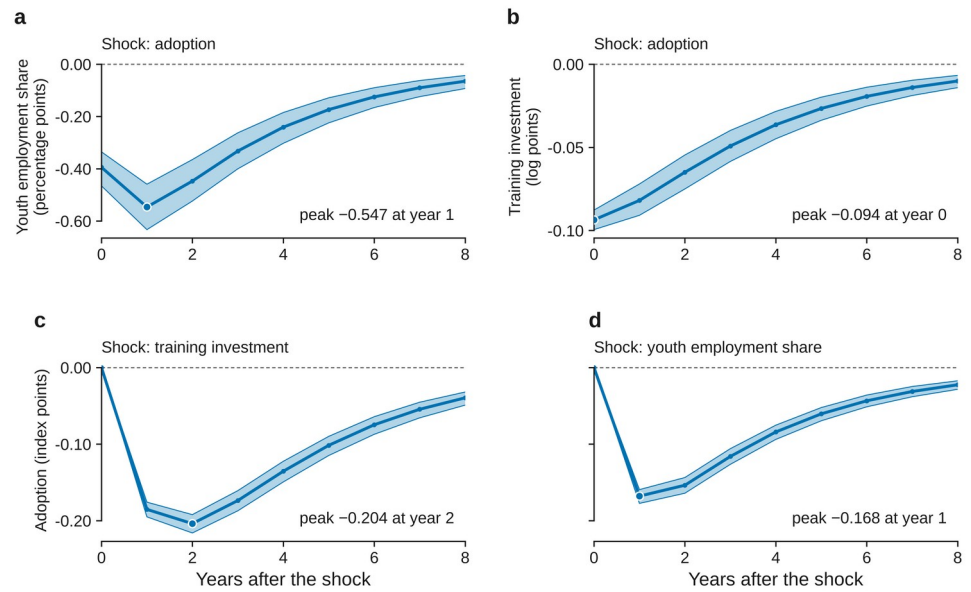


Figure 3. Orthogonalized impulse responses of the adoption-capability-youth system.

Table 12 converts the responses into the quantity that matters for the systems claim. Cumulated over eight years, the response of the youth employment share to an adoption innovation is -2.416 percentage points, 6.1 times the impact response. When the two return coefficients of the adoption equation are set to zero and the system is re-simulated, the same cumulative response is -2.030 . The ratio defined in Equation (9) is therefore 1.190: the reinforcing loop adds 19.0 percent to the eight-year cumulative displacement of young workers. The amplification is absent by construction in the first two years, because the return arc carries a one-period delay and the loop must travel the full circuit before it contributes, and it grows monotonically thereafter. This is the signature of a reinforcing feedback structure rather than of a persistent shock, and it is not recoverable from any single-equation estimate.

Table 12. Impulse responses and the amplification attributable to the reinforcing loop.

Response	h = 0	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8
Youth to a GenAI shock	−0.395 [−0.466, −0.336]	−0.547 [−0.633, −0.459]	−0.447 [−0.524, −0.365]	−0.332 [−0.400, −0.263]	−0.241 [−0.303, −0.185]	−0.174 [−0.225, −0.129]	−0.125 [−0.166, −0.090]	−0.090 [−0.124, −0.063]	−0.065 [−0.093, −0.043]
Train to a GenAI shock	−0.094 [−0.099, −0.088]	−0.082 [−0.091, −0.072]	−0.065 [−0.075, −0.055]	−0.049 [−0.058, −0.040]	−0.036 [−0.045, −0.028]	−0.027 [−0.034, −0.020]	−0.019 [−0.025, −0.014]	−0.014 [−0.019, −0.010]	−0.010 [−0.014, −0.007]
GenAI to a Train shock	0.000 [0.000, 0.000]	−0.185 [−0.195, −0.176]	−0.204 [−0.216, −0.192]	−0.174 [−0.187, −0.161]	−0.135 [−0.149, −0.123]	−0.102 [−0.115, −0.090]	−0.075 [−0.087, −0.064]	−0.055 [−0.066, −0.045]	−0.040 [−0.049, −0.032]
GenAI to a Youth shock	0.000 [0.000, 0.000]	−0.168 [−0.177, −0.159]	−0.154 [−0.164, −0.144]	−0.116 [−0.126, −0.106]	−0.084 [−0.094, −0.075]	−0.061 [−0.070, −0.052]	−0.043 [−0.051, −0.036]	−0.031 [−0.038, −0.025]	−0.023 [−0.028, −0.017]
Cumulative Youth response, closed loop	−0.395	−0.941	−1.389	−1.721	−1.962	−2.136	−2.261	−2.351	−2.416
Cumulative Youth response, loop suppressed	−0.395	−0.941	−1.360	−1.636	−1.808	−1.912	−1.973	−2.009	−2.030
Amplification factor	1.000	1.000	1.021	1.052	1.085	1.117	1.146	1.170	1.190

Note: Responses are orthogonalized by a Cholesky decomposition with the variables ordered GenAI, Train, Youth, so that adoption may move the other two contemporaneously but not the reverse. Figures in square brackets are 90% bootstrap intervals. The loop-suppressed path sets the two return coefficients of the GenAI equation to zero and re-simulates the system; the amplification factor is the ratio of the two cumulative paths. Source: authors' own calculations.

5. Discussion and Implications

5.1. The Displacement of Entry-Level Labor (H1)

Hypothesis 1 was confirmed. Generative artificial intelligence adoption reduces the youth employment share of the adopting firm, by 1.74 percentage points per standard deviation of adoption intensity, and the estimate survives the replacement of the year effects with industry-by-year and province-by-year effects, an overidentified instrumental-variable design, propensity score matching, entropy balancing and six robustness specifications. The result is consistent with the exposure evidence of Refs. [15] and [16] and with the aggregate relative decline in young workers' employment documented by Ref. [17], and it locates that decline inside the firm, where the hiring decision is actually taken. The placebo result is what gives the finding its interpretive force: the effect on employees aged 31 to 50 is 11.2 times smaller, so what the estimates capture is not a general contraction of employment but a change in its age composition.

5.2. Two Mechanisms of Displacement (H2 and H3)

Hypotheses 2 and 3 were both supported, and their relative sizes carry a message that neither hypothesis alone conveys. The contraction of the routine task base accounts for 22.3 percent of the total effect, which is the mechanism the task-based framework of Refs. [19] and [44] predicts and which most of the existing literature assumes to be the whole story. The suppression of training investment accounts for a further 12.5 percent, a channel that the task-based framework does not contain at all, because it treats labor as an input to be allocated rather than an asset to be produced. This second channel is the point at which the labor economics of automation meets the organizational learning tradition of Refs. [32] and [36]. A firm that reduces its training outlay when it adopts is not merely hiring fewer juniors this year; it is dismantling the apparatus by which it converts juniors into seniors, which is precisely the asset that Refs. [28] and [29] identified as unavailable on the external market.

5.3. What Weakens and What Sharpens the Effect (H4a and H4b)

Hypotheses 4a and 4b were both validated. The displacement is roughly halved in firms one standard deviation above the mean of organizational learning capability and correspondingly deepened in firms above the mean of relative labor cost, and the two conditions operate independently. This is the empirical content of the socio-technical proposition that the technical and social subsystems must be jointly optimized [39,40]: the same technology, adopted at the same measured intensity, produces materially different employment outcomes depending on the organizational capacity that receives it. The heterogeneity results in Section 4.7 reinforce the reading. Industry classification, which proxies for learning capability only loosely, does not discriminate between firms, whereas the direct measure does; and the labor-intensity split, which proxies for the size of the routine task base, discriminates sharply. The determinants of the outcome are firm-level and, in the case of learning capability, constructible.

5.4. The Reinforcing Loop and the Capability Trap (H5)

Hypothesis 5 is strongly supported by the system estimates, and it is the finding on which the contribution of this paper rests. The two return arcs are negative and significant: a firm whose training investment has fallen, and

whose youth cohort has thinned, adopts more generative artificial intelligence in the following year. The structure is exactly the capability trap described by Ref. [34] and formalized by Ref. [35]. A short-run fix and a long-run capability investment compete for the same managerial attention; the fix pays immediately and the investment pays late; choosing the fix degrades the capability whose weakness makes the fix attractive; and the pull toward the fix is therefore stronger next period than it was this period. No step in this sequence requires a mistake. The pathology is a property of the loop, not of the decision-maker, which is why the loop and not the decision is the object of intervention [70].

The quantitative content is that the loop adds 19.0 percent to the eight-year cumulative displacement. Two readings follow. The first is methodological: any estimate of the effect of generative artificial intelligence on youth employment obtained from a single equation understates the total effect by roughly this margin, because it holds constant a variable that the outcome itself moves. The second is substantive: because the largest eigenvalue is 0.722 rather than greater than one, the trap is bounded, the accumulation converges, and an intervention that cuts the return arc, rather than one that resists the initial adoption, returns the system to the smaller open-loop path. This is what a leverage point means in the systems sense: a place where a small structural change produces a large behavioral change, identified from the structure of the loop rather than from the size of any single coefficient [68,70,71].

5.5. Theoretical Contributions

This study makes three theoretical contributions. First, it reconceptualizes the relationship between generative artificial intelligence and youth employment as a feedback structure rather than a treatment effect, and supplies the first firm-level estimate of the return arc together with a measure of the amplification it produces. This extends the task-based framework of Refs. [19] and [20] in a direction that framework does not itself provide, because the framework has no representation of the firm's technology choice as a function of its own workforce composition. Second, it brings the capability trap of Refs. [34] and [35], previously demonstrated for process improvement and maintenance, to bear on human capital, and shows that the same structure governs the erosion of a firm's skill-formation system. Third, it operationalizes the socio-technical proposition of Refs. [39] and [40] in a form that can be estimated, by showing that the employment consequence of a fixed technical intensity varies systematically with a measurable property of the social subsystem.

5.6. Managerial and Policy Implications

For firm managers, the results argue against evaluating generative artificial intelligence deployments on the cost saving of the period in which they are made. The saving is real, and so is the erosion of the pipeline that produces the firm's future senior staff; the second is delayed and therefore systematically underweighted. Three concrete practices follow: ring-fence the training budget from the productivity savings that adoption generates, so that the two decisions are not made against one another; redesign entry-level roles around the complementary tasks that the technology creates, namely prompt design, output verification, data curation and workflow integration, rather than eliminating the roles whose original tasks the technology absorbs; and

monitor the age structure of the workforce as a leading indicator of capability, not as a by-product of hiring.

For human resource and workforce development functions, the moderation results identify organizational learning capability as the leverage point. Because the same adoption intensity displaces roughly half as much youth employment in firms with strong absorptive capacity, investment in that capacity is not a competing use of funds but the condition under which adoption and youth employment cease to be substitutes.

For policymakers, the implications are addressed to specific actors. Employment ministries and local labor bureaux should recognize that subsidies attached to the act of hiring a graduate operate on the forward arc of the loop and will be partly undone by the return arc; instruments that condition support on verified training provision, rather than on headcount alone, act on the arc that closes the loop and are correspondingly more durable. Industrial and technology regulators implementing national artificial-intelligence strategies [100,101] should incorporate workforce development conditions into deployment incentives, since the evidence here indicates that adoption and skill formation are not automatically complementary at the firm level. Education authorities facing record graduate cohorts [2] should note that the tasks disappearing from entry-level roles are the codifiable ones, so curricula that certify verification, judgement and model-integration skills address the margin that is actually moving. Finally, state-owned enterprises, whose displacement coefficient is the smallest in Table 10, are already functioning as a partial buffer; the finding argues for making that role explicit and time-limited rather than incidental, since a buffer that is not accompanied by training provision postpones the trap without disarming it.

5.7. Limitations and Future Research Directions

Several limitations qualify these findings and point to further work. First, adoption is measured from annual-report text, which captures the intensity with which a firm reports applying the technology rather than verified deployment; firms may over-report to signal modernity, and although firm fixed effects absorb persistent differences in disclosure style, time-varying strategic disclosure would remain. Linking the text measure to procurement records or to model-usage telemetry would sharpen the measurement. Second, the sample covers listed firms, which are larger and more formal than the enterprises that employ most young Chinese workers; the direction of the bias is not obvious, since listed firms adopt earlier but also carry more structured internal labor markets, and extending the design to survey data on small and medium enterprises would settle it. Third, the panel covers 9 years, of which only the last two lie after the generative-AI break, so the impulse responses beyond a three-year horizon are extrapolations of the estimated dynamics rather than direct observations; re-estimating the system as the post-break window lengthens is the natural next step. Fourth, the analysis stops at the firm boundary and therefore cannot say whether the young workers not hired by adopting firms are absorbed elsewhere in the economy; combining firm-level adoption with matched employer-employee records would identify the destination of the displaced. Fifth, the identification of the return arc rests on a recursive ordering of the system, and although the ordering is the conventional one for a technology variable that is decided

before the annual workforce plan is executed, a design with an external shifter of training investment would provide a stronger test. Finally, the mediators measured here do not exhaust the mechanism; organizational structure, supervisory capacity and the internal promotion ladder are plausible additional channels that richer personnel data would allow future research to examine.

6. Conclusions

Entry-level work is the first rung of the ladder by which economies produce experienced workers, and generative artificial intelligence is most capable at exactly the tasks that rung is made of. Using 25,701 firm-year observations for 3,000 Chinese A-share listed firms over 2016–2024, this study finds that a one-standard-deviation increase in generative-AI application intensity lowers the youth employment share by 1.74 percentage points, or 6.0 percent of its mean, while employment of workers aged 31 to 50 is barely affected. The displacement operates through two measurable channels: a contracting routine task base, which accounts for 22.3 percent of the total effect, and suppressed investment in employee training, which accounts for a further 12.5 percent. It is weaker where organizational learning capability is stronger and sharper where relative labor costs are higher, and it is about a quarter larger after the generative-AI break of late 2022 than before it.

The central finding is structural rather than marginal. The adoption-capability-youth relationship is not a one-way shock but a reinforcing loop: firms whose training investment and youth cohort have contracted adopt more generative artificial intelligence in the following year, and this return arc amplifies the eight-year cumulative displacement by 19.0 percent relative to a system in which the loop is switched off. Because the largest eigenvalue of the estimated system is 0.722, the trap is bounded rather than explosive, which is what makes it addressable. The policy conclusion follows from the structure: interventions aimed at the moment of adoption operate on the forward arc and are partly undone by the return arc, whereas instruments that protect training provision act on the arc that closes the loop. For a country entering a decade of record graduate cohorts, the difference between the two is the difference between slowing the erosion of the first rung and stopping it.

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References

1. International Labour Organization. *World Employment and Social Outlook: Trends 2025*; International Labour Office: Geneva, Switzerland, 2025.
2. Ministry of Education of the People's Republic of China. The Number of College Graduates Nationwide Is Expected to Reach 12.22 Million in 2025. Available online: <http://www.moe.gov.cn/> (accessed on 20 March 2026).
3. National Bureau of Statistics of China. Surveyed Urban Unemployment Rate of the Population Aged 16 to 24 Excluding Students. Available online: <https://www.stats.gov.cn/english/> (accessed on 20 March 2026).
4. International Labour Organization. *Global Employment Trends for Youth 2024: Decent Work, Brighter Futures*; International Labour Office: Geneva, Switzerland, 2024.
5. World Economic Forum. *The Future of Jobs Report 2025*; World Economic Forum: Geneva, Switzerland, 2025.
6. Kahn, L.B. The Long-Term Labor Market Consequences of Graduating from College in a Bad Economy. *Labour Econ.* **2010**, *17*, 303–316. [CrossRef]
7. Oreopoulos, P.; von Wachter, T.; Heisz, A. The Short- and Long-Term Career Effects of Graduating in a Recession. *Am. Econ. J. Appl. Econ.* **2012**, *4*, 1–29. [CrossRef]
8. Schwanndt, H.; von Wachter, T. Unlucky Cohorts: Estimating the Long-Term Effects of Entering the Labor Market in a Recession in Large Cross-Sectional Data Sets. *J. Labor Econ.* **2019**, *37*, S161–S198. [CrossRef]
9. von Wachter, T. The Persistent Effects of Initial Labor Market Conditions for Young Adults and Their Sources. *J. Econ. Perspect.* **2020**, *34*, 168–194. [CrossRef]
10. Bick, A.; Blandin, A.; Deming, D.J. *The Rapid Adoption of Generative AI*; NBER Working Paper 32966; National Bureau of Economic Research: Cambridge, MA, USA, 2024.
11. Brynjolfsson, E.; Li, D.; Raymond, L.R. Generative AI at Work. *Q. J. Econ.* **2025**, *140*, 889–942. [CrossRef]
12. Noy, S.; Zhang, W. Experimental Evidence on the Productivity Effects of Generative Artificial Intelligence. *Science* **2023**, *381*, 187–192. [CrossRef]
13. Dell'Acqua, F.; McFowland, E., III; Mollick, E.R.; Lifshitz-Assaf, H.; Kellogg, K.C.; Rajendran, S.; Kraymer, L.; Candelon, F.; Lakhani, K.R. Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of Artificial Intelligence on Knowledge Worker Productivity and Quality. *Organ. Sci.* **2026**, in press. [CrossRef]
14. Peng, S.; Kalliamvakou, E.; Cihon, P.; Demirer, M. The Impact of AI on Developer Productivity: Evidence from GitHub Copilot. *arXiv* **2023**, arXiv:2302.06590.
15. Eloundou, T.; Manning, S.; Mishkin, P.; Rock, D. GPTs Are GPTs: Labor Market Impact Potential of LLMs. *Science* **2024**, *384*, 1306–1308. [CrossRef]
16. Felten, E.; Raj, M.; Seamans, R. Occupational, Industry, and Geographic Exposure to Artificial Intelligence: A Novel Dataset and Its Potential Uses. *Strateg. Manag. J.* **2021**, *42*, 2195–2217. [CrossRef]
17. Brynjolfsson, E.; Chandar, B.; Chen, R. *Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence*; Working Paper; Stanford Digital Economy Lab, Stanford University: Stanford, CA, USA, 2025.
18. Liu, Y.; Wang, H.; Yu, S. *Labor Demand in the Age of Generative AI: Early Evidence from the U.S. Job Posting Data*; Policy Research Working Paper 11263; World Bank: Washington, DC, USA, 2025.
19. Acemoglu, D.; Restrepo, P. The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment. *Am. Econ. Rev.* **2018**, *108*, 1488–1542. [CrossRef]
20. Acemoglu, D.; Restrepo, P. Tasks, Automation, and the Rise in U.S. Wage Inequality. *Econometrica* **2022**, *90*, 1973–2016. [CrossRef]
21. Autor, D.; Chin, C.; Salomons, A.; Seegmiller, B. New Frontiers: The Origins and Content of New Work, 1940–2018. *Q. J. Econ.* **2024**, *139*, 1399–1465. [CrossRef]
22. Bresnahan, T.F.; Trajtenberg, M. General Purpose Technologies: “Engines of Growth”? *J. Econom.* **1995**, *65*, 83–108. [CrossRef]
23. Agrawal, A.; Gans, J.S.; Goldfarb, A. Artificial Intelligence: The Ambiguous Labor Market Impact of Automating Prediction. *J. Econ. Perspect.* **2019**, *33*, 31–50. [CrossRef]
24. Brynjolfsson, E.; Rock, D.; Syverson, C. The Productivity J-Curve: How Intangibles Complement General Purpose Technologies. *Am. Econ. J. Macroecon.* **2021**, *13*, 333–372. [CrossRef]
25. Babina, T.; Fedyk, A.; He, A.; Hodson, J. Artificial Intelligence, Firm Growth, and Product Innovation. *J. Financ. Econ.* **2024**, *151*, 103745. [CrossRef]
26. Hui, X.; Reshef, O.; Zhou, L. The Short-Term Effects of Generative Artificial Intelligence on Employment: Evidence from an Online Labor Market. *Organ. Sci.* **2024**, *35*, 1977–1989. [CrossRef]
27. Humlum, A.; Vestergaard, E. *Large Language Models, Small Labor Market Effects*; NBER Working Paper 33777; National Bureau of Economic Research: Cambridge, MA, USA, 2025.
28. Becker, G.S. *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*; National Bureau of Economic Research: New York, NY, USA, 1964.

- 1048 29. Doeringer, P.B.; Piore, M.J. *Internal Labor Markets and Manpower Analysis*; D.C. Heath: Lexington, MA, USA, 1971.
- 1049
- 1050 30. Gibbons, R.; Waldman, M. Task-Specific Human Capital. *Am. Econ. Rev.* **2004**, *94*, 203–207. [CrossRef]
- 1051 31. Acemoglu, D.; Pischke, J.-S. Why Do Firms Train? Theory and Evidence. *Q. J. Econ.* **1998**, *113*, 79–119.
- 1052 [CrossRef]
- 1053 32. Cohen, W.M.; Levinthal, D.A. Absorptive Capacity: A New Perspective on Learning and Innovation. *Adm. Sci. Q.* **1990**, *35*, 128–152. [CrossRef]
- 1054
- 1055 33. Argote, L.; Miron-Spektor, E. Organizational Learning: From Experience to Knowledge. *Organ. Sci.* **2011**, *22*, 1123–1137. [CrossRef]
- 1056
- 1057 34. Repenning, N.P.; Sterman, J.D. Capability Traps and Self-Confirming Attribution Errors in the Dynamics of Process Improvement. *Adm. Sci. Q.* **2002**, *47*, 265–295. [CrossRef]
- 1058
- 1059 35. Rahmandad, H.; Repenning, N. Capability Erosion Dynamics. *Strateg. Manag. J.* **2016**, *37*, 649–672. [CrossRef]
- 1060 36. Levinthal, D.A.; March, J.G. The Myopia of Learning. *Strateg. Manag. J.* **1993**, *14*, 95–112. [CrossRef]
- 1061 37. von Bertalanffy, L. *General System Theory: Foundations, Development, Applications*; George Braziller: New York, NY, USA, 1968.
- 1062
- 1063 38. Holland, J.H. *Hidden Order: How Adaptation Builds Complexity*; Addison-Wesley: Reading, MA, USA, 1995.
- 1064 39. Trist, E.L.; Bamforth, K.W. Some Social and Psychological Consequences of the Longwall Method of Coal-Getting. *Hum. Relat.* **1951**, *4*, 3–38. [CrossRef]
- 1065
- 1066 40. Emery, F.E.; Trist, E.L. The Causal Texture of Organizational Environments. *Hum. Relat.* **1965**, *18*, 21–32. [CrossRef]
- 1067
- 1068 41. Autor, D.H. Why Are There Still So Many Jobs? The History and Future of Workplace Automation. *J. Econ. Perspect.* **2015**, *29*, 3–30. [CrossRef]
- 1069
- 1070 42. Acemoglu, D.; Restrepo, P. Robots and Jobs: Evidence from US Labor Markets. *J. Political Econ.* **2020**, *128*, 2188–2244. [CrossRef]
- 1071
- 1072 43. Bonfiglioli, A.; Crinò, R.; Fadinger, H.; Gancia, G. Robot Imports and Firm-Level Outcomes. *Econ. J.* **2024**, *134*, 3428–3444. [CrossRef]
- 1073
- 1074 44. Autor, D.H.; Levy, F.; Murnane, R.J. The Skill Content of Recent Technological Change: An Empirical Exploration. *Q. J. Econ.* **2003**, *118*, 1279–1333. [CrossRef]
- 1075
- 1076 45. Beane, M. Shadow Learning: Building Robotic Surgical Skill When Approved Means Fail. *Adm. Sci. Q.* **2019**, *64*, 87–123. [CrossRef]
- 1077
- 1078 46. Kellogg, K.C.; Valentine, M.A.; Christin, A. Algorithms at Work: The New Contested Terrain of Control. *Acad. Manag. Ann.* **2020**, *14*, 366–410. [CrossRef]
- 1079
- 1080 47. Cui, Z.K.; Demirer, M.; Jaffe, S.; Musolff, L.; Peng, S.; Salz, T. The Effects of Generative AI on High-Skilled Work: Evidence from Three Field Experiments with Software Developers. *Manag. Sci.* **2026**, in press. [CrossRef]
- 1081
- 1082
- 1083 48. Otis, N.G.; Clarke, R.; Delecourt, S.; Holtz, D.; Koning, R. The Uneven Impact of Generative Artificial Intelligence on Entrepreneurial Performance: Evidence from a Field Experiment in Kenya. *Manag. Sci.* **2026**, in press. [CrossRef]
- 1084
- 1085
- 1086 49. Raisch, S.; Krakowski, S. Artificial Intelligence and Management: The Automation-Augmentation Paradox. *Acad. Manag. Rev.* **2021**, *46*, 192–210. [CrossRef]
- 1087
- 1088 50. Krakowski, S.; Luger, J.; Raisch, S. Artificial Intelligence and the Changing Sources of Competitive Advantage. *Strateg. Manag. J.* **2023**, *44*, 1425–1452. [CrossRef]
- 1089
- 1090 51. Forsythe, E. Why Don't Firms Hire Young Workers during Recessions? *Econ. J.* **2022**, *132*, 1765–1789. [CrossRef]
- 1091
- 1092 52. Xue, F.; Jin, S. Artificial Intelligence Adoption, Innovation Efficiency, and Governance Mechanisms: Evidence from China. *Systems* **2025**, *13*, 1062. [CrossRef]
- 1093
- 1094 53. Xue, X.; Li, L.; Chen, J.; Luo, T. How Does Digital Technology Innovation Quality Empower Corporate ESG Performance? The Roles of Digital Transformation and Digital Technology Diffusion. *Systems* **2025**, *13*, 929. [CrossRef]
- 1095
- 1096
- 1097 54. Machucho, R.; Ortiz, D. The Impacts of Artificial Intelligence on Business Innovation: A Comprehensive Review of Applications, Organizational Challenges, and Ethical Considerations. *Systems* **2025**, *13*, 264. [CrossRef]
- 1098
- 1099 55. Liang, H.; Fan, J.; Wang, Y. Artificial Intelligence, Technological Innovation, and Employment Transformation for Sustainable Development: Evidence from China. *Sustainability* **2025**, *17*, 3842. [CrossRef]
- 1100
- 1101 56. Acemoglu, D.; Pischke, J.-S. Beyond Becker: Training in Imperfect Labour Markets. *Econ. J.* **1999**, *109*, F112–F142. [CrossRef]
- 1102
- 1103 57. Lazear, E.P. Firm-Specific Human Capital: A Skill-Weights Approach. *J. Political Econ.* **2009**, *117*, 914–940. [CrossRef]
- 1104
- 1105 58. March, J.G. Exploration and Exploitation in Organizational Learning. *Organ. Sci.* **1991**, *2*, 71–87. [CrossRef]
- 1106 59. Yang, B.; Sun, Y.; Zeng, Z.; Li, Q. Deskillling, Reskillling, or Upskillling? Unpacking the Pathways of Student Adaptation to Generative Artificial Intelligence. *Int. J. Inf. Manag.* **2026**, *87*, 103002. [CrossRef]
- 1107
- 1108 60. Arrow, K.J. The Economic Implications of Learning by Doing. *Rev. Econ. Stud.* **1962**, *29*, 155–173. [CrossRef]

61. Nelson, R.R.; Winter, S.G. *An Evolutionary Theory of Economic Change*; Harvard University Press: Cambridge, MA, USA, 1982.
62. Acemoglu, D.; Autor, D.; Hazell, J.; Restrepo, P. Artificial Intelligence and Jobs: Evidence from Online Vacancies. *J. Labor Econ.* **2022**, *40*, S293–S340. [CrossRef]
63. Checkland, P. *Systems Thinking, Systems Practice*; Wiley: Chichester, UK, 1981.
64. Teece, D.J.; Pisano, G.; Shuen, A. Dynamic Capabilities and Strategic Management. *Strateg. Manag. J.* **1997**, *18*, 509–533. [CrossRef]
65. Autor, D. *Applying AI to Rebuild Middle Class Jobs*; NBER Working Paper 32140; National Bureau of Economic Research: Cambridge, MA, USA, 2024.
66. Vaccaro, M.; Almaatouq, A.; Malone, T. When Combinations of Humans and AI Are Useful: A Systematic Review and Meta-Analysis. *Nat. Hum. Behav.* **2024**, *8*, 2293–2303. [CrossRef]
67. Bloom, N.; Van Reenen, J. Measuring and Explaining Management Practices across Firms and Countries. *Q. J. Econ.* **2007**, *122*, 1351–1408. [CrossRef]
68. Sterman, J.D. *Business Dynamics: Systems Thinking and Modeling for a Complex World*; Irwin/McGraw-Hill: Boston, MA, USA, 2000.
69. Forrester, J.W. *Industrial Dynamics*; MIT Press: Cambridge, MA, USA, 1961.
70. Meadows, D.H. *Thinking in Systems: A Primer*; Chelsea Green Publishing: White River Junction, VT, USA, 2008.
71. Senge, P.M. *The Fifth Discipline: The Art and Practice of the Learning Organization*; Doubleday: New York, NY, USA, 1990.
72. Sims, C.A. Macroeconomics and Reality. *Econometrica* **1980**, *48*, 1–48. [CrossRef]
73. Love, I.; Zicchino, L. Financial Development and Dynamic Investment Behavior: Evidence from Panel VAR. *Q. Rev. Econ. Finance* **2006**, *46*, 190–210. [CrossRef]
74. Ashby, W.R. *An Introduction to Cybernetics*; Chapman and Hall: London, UK, 1956.
75. Wu, F.; Hu, H.; Lin, H.; Ren, X. Enterprise Digital Transformation and Capital Market Performance: Empirical Evidence from Stock Liquidity. *J. Manag. World* **2021**, *37*, 130–144.
76. Yuan, C.; Xiao, S.; Geng, C.; Sheng, Y. Digital Transformation and Division of Labour between Enterprises: Vertical Specialisation or Vertical Integration. *China Ind. Econ.* **2021**, *9*, 137–155.
77. Zhao, C.; Wang, W.; Li, X. How Does Digital Transformation Affect the Total Factor Productivity of Enterprises? *Financ. Trade Econ.* **2021**, *42*, 114–129.
78. Aiken, L.S.; West, S.G. *Multiple Regression: Testing and Interpreting Interactions*; Sage: Newbury Park, CA, USA, 1991.
79. Petersen, M.A. Estimating Standard Errors in Finance Panel Data Sets: Comparing Approaches. *Rev. Financ. Stud.* **2009**, *22*, 435–480. [CrossRef]
80. Abadie, A.; Athey, S.; Imbens, G.W.; Wooldridge, J.M. When Should You Adjust Standard Errors for Clustering? *Q. J. Econ.* **2023**, *138*, 1–35. [CrossRef]
81. Baron, R.M.; Kenny, D.A. The Moderator-Mediator Variable Distinction in Social Psychological Research: Conceptual, Strategic, and Statistical Considerations. *J. Pers. Soc. Psychol.* **1986**, *51*, 1173–1182. [CrossRef]
82. Zhao, X.; Lynch, J.G., Jr.; Chen, Q. Reconsidering Baron and Kenny: Myths and Truths about Mediation Analysis. *J. Consum. Res.* **2010**, *37*, 197–206. [CrossRef]
83. Preacher, K.J.; Hayes, A.F. Asymptotic and Resampling Strategies for Assessing and Comparing Indirect Effects in Multiple Mediator Models. *Behav. Res. Methods* **2008**, *40*, 879–891. [CrossRef]
84. Kleibergen, F.; Paap, R. Generalized Reduced Rank Tests Using the Singular Value Decomposition. *J. Econom.* **2006**, *133*, 97–126. [CrossRef]
85. Stock, J.H.; Yogo, M. Testing for Weak Instruments in Linear IV Regression. In *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg*; Andrews, D.W.K., Stock, J.H., Eds.; Cambridge University Press: Cambridge, UK, 2005; pp. 80–108.
86. Young, A. Consistency without Inference: Instrumental Variables in Practical Application. *Eur. Econ. Rev.* **2022**, *147*, 104112. [CrossRef]
87. Wooldridge, J.M. *Econometric Analysis of Cross Section and Panel Data, 2nd ed.*; MIT Press: Cambridge, MA, USA, 2010.
88. Holtz-Eakin, D.; Newey, W.; Rosen, H.S. Estimating Vector Autoregressions with Panel Data. *Econometrica* **1988**, *56*, 1371–1395. [CrossRef]
89. Abrigo, M.R.M.; Love, I. Estimation of Panel Vector Autoregression in Stata. *Stata J.* **2016**, *16*, 778–804. [CrossRef]
90. Nickell, S. Biases in Dynamic Models with Fixed Effects. *Econometrica* **1981**, *49*, 1417–1426. [CrossRef]
91. Arellano, M.; Bond, S. Some Tests of Specification for Panel Data: Monte Carlo Evidence and an Application to Employment Equations. *Rev. Econ. Stud.* **1991**, *58*, 277–297. [CrossRef]
92. Arellano, M.; Bover, O. Another Look at the Instrumental Variable Estimation of Error-Components Models. *J. Econom.* **1995**, *68*, 29–51. [CrossRef]
93. Blundell, R.; Bond, S. Initial Conditions and Moment Restrictions in Dynamic Panel Data Models. *J. Econom.* **1998**, *87*, 115–143. [CrossRef]

- 1170 94. Chudik, A.; Pesaran, M.H.; Yang, J.-C. Half-Panel Jackknife Fixed-Effects Estimation of Linear Models with
1171 Weakly Exogenous Regressors. *J. Appl. Econom.* **2018**, *33*, 816–836. [CrossRef]
- 1172 95. Rosenbaum, P.R.; Rubin, D.B. The Central Role of the Propensity Score in Observational Studies for Causal
1173 Effects. *Biometrika* **1983**, *70*, 41–55. [CrossRef]
- 1174 96. Oster, E. Unobservable Selection and Coefficient Stability: Theory and Evidence. *J. Bus. Econ. Stat.* **2019**, *37*,
1175 187–204. [CrossRef]
- 1176 97. Cameron, A.C.; Gelbach, J.B.; Miller, D.L. Bootstrap-Based Improvements for Inference with Clustered Errors.
1177 *Rev. Econ. Stat.* **2008**, *90*, 414–427. [CrossRef]
- 1178 98. Roodman, D.; Nielsen, M.Ø.; MacKinnon, J.G.; Webb, M.D. Fast and Wild: Bootstrap Inference in Stata Using
1179 boottest. *Stata J.* **2019**, *19*, 4–60. [CrossRef]
- 1180 99. Dwivedi, Y.K.; Kshetri, N.; Hughes, L.; Slade, E.L.; Jeyaraj, A.; Kar, A.K.; Baabdullah, A.M.; Koohang, A.;
1181 Raghavan, V.; Ahuja, M.; et al. Opinion Paper: “So What If ChatGPT Wrote It?” Multidisciplinary Perspectives
1182 on Opportunities, Challenges and Implications of Generative Conversational AI for Research, Practice and
1183 Policy. *Int. J. Inf. Manag.* **2023**, *71*, 102642. [CrossRef]
- 1184 100. State Council of the People’s Republic of China. Opinions on Deepening the Implementation of the “Artificial
1185 Intelligence Plus” Initiative. Available online: <https://www.gov.cn/> (accessed on 20 March 2026).
- 1186 101. Janssen, M. Responsible Governance of Generative AI: Conceptualizing GenAI as Complex Adaptive Systems.
1187 *Policy Soc.* **2025**, *44*, 38–51. [CrossRef]